

Prof. Dr.-Ing. Viktor Wesselak

Writing a poor thesis - a bad-practice-guide

Lecture 7 - Bad Practice

Caution: This presentation may contain irony!

Equations

Genauigkeit von U_b durch iterative Anwendung folgender Formel erreicht werden [9]:

$$U \approx U_b + P_b f \quad \text{mit} \quad f = \frac{1 - [1 - (\ln(1 - I_{nb} / I) / B_e)]}{(2 I_{nb} / B_e)} \quad (3.1)$$

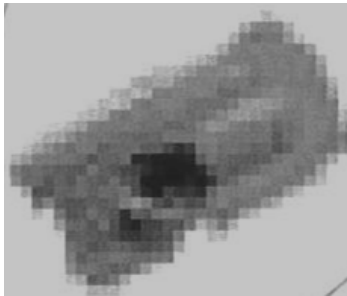
Did you write important formulas much larger than the rest so that the reader realizes that something important is happening here?

$$P_V = \frac{(3230W)^2 * 2 * 35m}{6mm^2 * 57 \frac{m}{\Omega * mm^2} * (437V)^2} = 11,18W$$

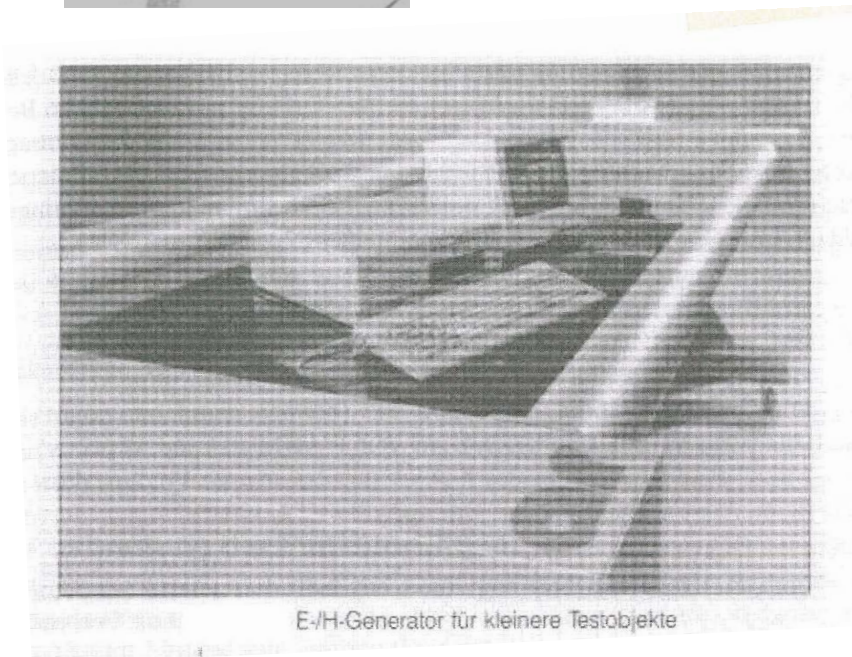
Do you think that a difficult equation looks friendlier if you decorate it with flowers?

Remember: a star is the symbol for the mathematical convolution. And by the way: variables are printed cursive – units never.

Figures



Are you trying to keep your document manageably small when saving it, if you don't use a high image resolution?



Are you trying to give your work a unique artistic signature by scanning low resolution images several times?

By the way: this is called moiré-effect and is caused by a violation of the Nyquist–Shannon sampling theorem.

Figures

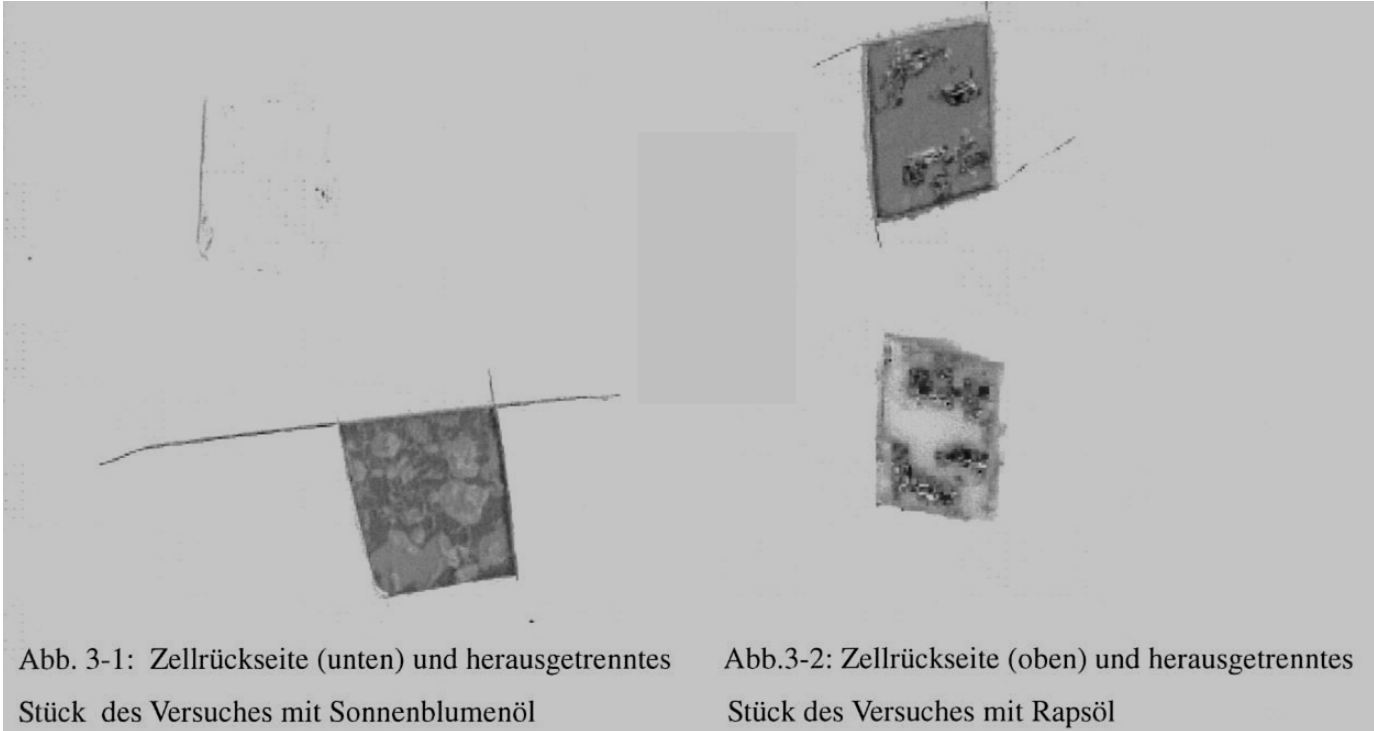
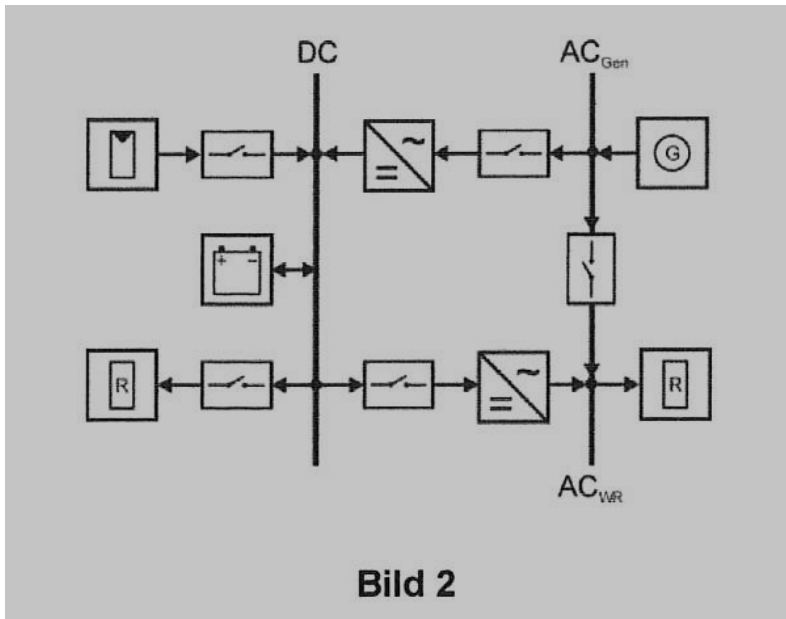


Abb. 3-1: Zellrückseite (unten) und herausgetrenntes Stück des Versuches mit Sonnenblumenöl

Abb.3-2: Zellrückseite (oben) und herausgetrenntes Stück des Versuches mit Rapsöl

Flying toast slices? I don't know, especially not how to rate it.

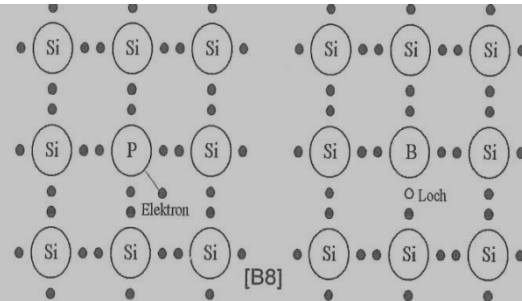
Figures



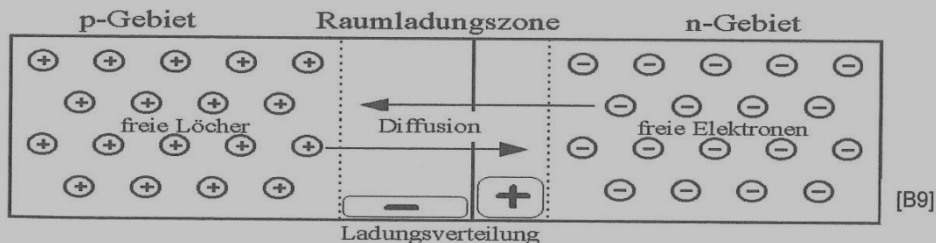
Okay, a picture is worth more than 1000 words, but that's why you don't use captions at all?

Figures

Dotiert man das Silizium mit Elementen aus der III. Hauptgruppe wie beispielsweise Bor (B) so fehlt ein Valenzelektron und es entsteht ein „Loch“ als Störstelle, welches durch geringen Energieeinsatz zum „wandern“ gebracht werden kann. Wird es mit einem Element der V. Hauptgruppe wie Phosphor (P) dotiert, so bleibt ein Elektron übrig, welches durch einen geringen Energieaufwand vom Atom gelöst werden kann und nun als freier Ladungsträger zur Verfügung steht. [3]



Bringt man nun diese beiden dotierten Halbleiter zusammen so entsteht ein p-n-Übergang. Da im n-dotierten Bereich ein Überschuss an Elektronen und im p-dotierten Bereich ein Überschuss an Löchern existiert wird es zu einem Konzentrationsausgleich kommen.



Das bedeutet, dass Elektronen in das p-Gebiet und Löcher in das n-Gebiet diffundieren. Wodurch ionisierte Dotierungsatome, welche fest in dem Kristallgitter eingebettet sind zurück bleiben. Diese bilden ein elektrisches Feld aus, welches dem Diffusionsstrom entgegen wirkt. Somit kann die Diffusion nicht endlos stattfinden.

Block typesetting really looks better, but is that why you need to make all the images wider?

Figures

CAPACITY AT VARIOUS DISCHARGE RATES (percentage of 20-hour rate)	
Hours to Discharge	Capacity (percent of rating)
20	100%
10	89%
5	78%
3	66%
1	45%



If you copy even the simplest tables, people will think you are lazy rather than accurate ...

Spelling

that was apparently due to the lack of competence during installation. Also the above mentioned communication between the engineers of SKYTRON Energy and those of ECOSOL left much to be desired. The ECOSOL engineers were in possession of the installation manual of the monitoring system but no consultation about the exact installation was made.

Not good at spelling but very consistent ...

By the way: if you ignore the spelling check of MS-Word or similar, then I think you want to provoke me.

Objectivity

At the same time, City Solar invested a large part of its resources in its own research department. To this end, the company secured the know-how of renowned experts from the industry early on in order to enrich the market with new technologies [1].

There should be a difference between science and advertising.

The CO₂ released during combustion changes our climate. The polar caps are melting. The sea level is rising. The deserts are constantly growing. Storms devastate entire regions. And there is still no turning point in sight. Politicians argue and take pro and contra positions, depending on their economic interests (USA).

And - of course - there should be a difference between science and opinion!

Citing

If you want to find out more about the production and structuring steps, I recommend the articles on this in the specialist book "Photovoltaische Anlagen" published by the German Solar Energy Society. Here the procedure is described very well.

Nice try for a final thesis.

Citing

Global energy consumption has increased dramatically over the last century. Due to the beginning of the industrial revolution in the 19th century, the development of the automobile and not least the constant industrialisation of cities, the demand for fossil fuels has almost exploded. According to estimates, almost all fossil energy resources will be used up by 2050. ¹⁾

¹⁾ www.evangelisches-gymnasium-meinerzhagen.de/projekte/2005/

It's nice if you also give sources from the 1534th page of the google hit list a chance. But perhaps there are one or two better sources for the problem of global resources than the project work of an grammar school.

Citing

that is not necessary. For 10 years, solar grade silicon (SG-Si) with a lower degree of impurity of about 0.01% has been used. As a result of this change, the PV modules produced have become more cost-effective. ³⁾

(...)

²⁾ [GH-2005]

³⁾ [DGS-1981]

I accept that the simple fact that a statement is 35 years old does not make it false.

Plagiarism



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Die freie Enzyklopädie

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Absorptionskältemaschine

Eine **Absorptionskältemaschine** (kurz AKM) ist eine [Kältemaschine](#), bei der im Gegensatz zur [Kompressionskältemaschine](#) die Verdichtung durch eine temperaturbeeinflusste Lösung des [Kältemittels](#) erfolgt. Man bezeichnet dies auch als **thermischen Verdichter**. Das Kältemittel wird in einem Lösungsmittelkreislauf bei geringer Temperatur in einem zweiten Stoff [absorbiert](#) und bei höheren Temperaturen [desorbiert](#). Bei dem Prozess wird die Löslichkeit zweier Stoffe genutzt. Die Stoffe in dem verwendeten Lösungsmittel sind.

Inhaltsverzeichnis

- [1 Entwicklung](#)
- [2 Grundlegende Prozesse](#)
- [3 Wärmeverhältnis](#)
- [4 Ammoniak-Wasser-Absorption](#)

3.5.4 Absorptionskältemaschine

Eine Absorptionskältemaschine ist eine Kältemaschine, bei der im Gegensatz zur Kompressionskältemaschine die Verdichtung durch eine temperaturbeeinflusste Lösung des Kältemittels erfolgt. Man bezeichnet dies auch als thermischen Verdichter. Das Kältemittel wird in einem Lösungsmittelkreislauf bei geringer Temperatur in einem zweiten Stoff absorbiert und bei höheren Temperaturen desorbiert. Bei dem Prozess wird die temperaturabhängige physikalische Löslichkeit von zwei Stoffen angewendet. Voraussetzung für den Prozess ist, dass die beiden Stoffe in dem verwendeten Temperaturintervall in jedem Verhältnis ineinander löslich sind.

Sometimes it's like this: you read a text and you know: I couldn't have expressed it better ...

Plagiarism

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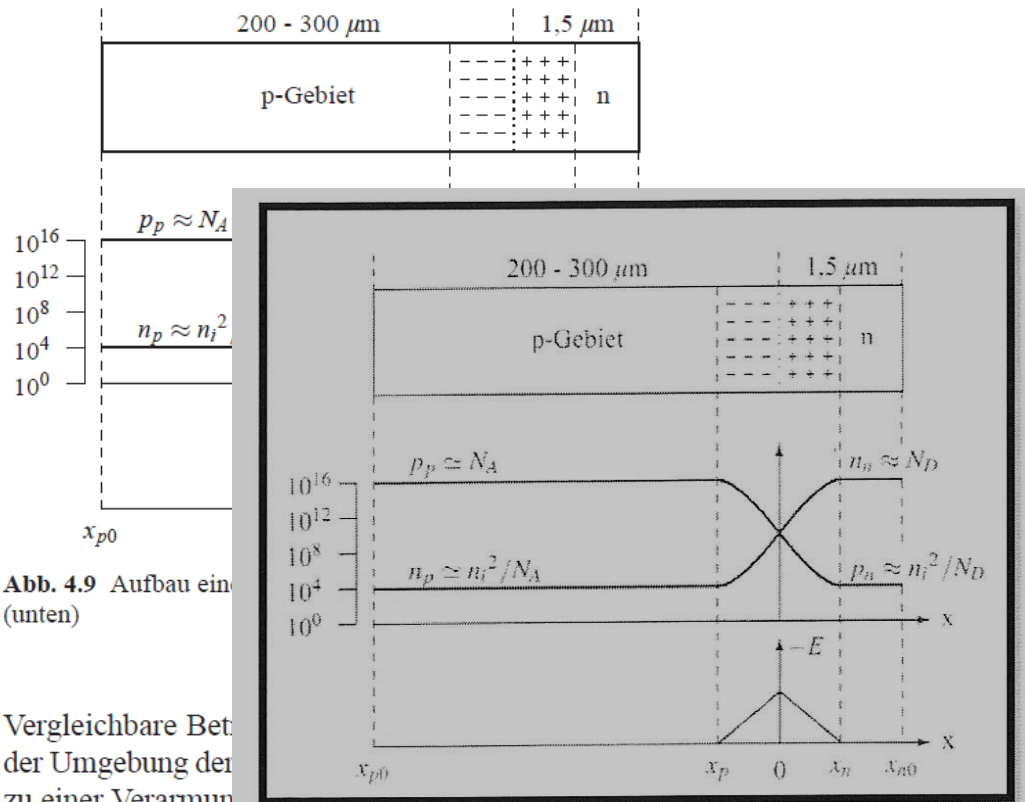


Abb. 4.9 Aufbau eines p-n-Übergangs (unten)

Vergleichbare Beträge der Umgebungs- und der Verarmungskonzentrationen



For risky people:
using a picture of
the supervisor but
not quoting.

Abb.2: Oben: Aufbau eines p-n-Übergangs
Mitte: Ladungsträgerdichte
Unten: elektrisches Feld

Plagiarism

Plagiarism seems to be a big problem for RES-students. Let's talk about it: Plagiarism is only one form of scientific misconduct. Other forms are:

- falsification of data or measurement results
- exploitation of dependency relationships
- deleting raw data

Plagiarism is a comparatively subordinate scientific misconduct and occurs predominantly in the text-intensive social sciences.

Plagiarism

Definition by Teddi Fishman [1]

- Uses words, ideas, or work products
- attributable to another identifiable person or source
- without attributing the work to the source from which it was obtained
- in a situation in which there is a legitimate expectation of original authorship
- in order to obtain some benefit, credit, or gain which need not be monetary

[1] Proceedings of the 4th Asia Pacific Conference on Educational Integrity. p.5

Plagiarism

Forms of academic plagiarism

- Submitting someone's work as their own.
- Re-writing someone's work without properly citing sources.
- Using quotations but not citing the source.
- Interweaving various sources together in the work without citing.
- References at a passage which conceals the scope of the takeover
- Providing proper citations, but failing to change the structure and wording of the borrowed ideas enough (close paraphrasing).
- Translation of a statement without citing the source
- Taking passages from their own previous work without adding citations (self-plagiarism).
- Relying too heavily on other people's work, failing to bring original thought into the text.

Chapter-2

Introduction to Urbanization

Urbanization refers to the population shift from rural areas to urban areas, the decrease in the proportion of people living in rural areas, and the ways in which each society adapts to this change.

Urban population is growing more rapidly than the rural, resulting gradual decrease in the percentage share of rural population among total population across the world. This urbanization process has been aggravated by the urban growth of Asian continent which has a lion- share of world population.

The South Asian region though predominantly rural (accounted for 69.9% rural population as on 2010), has recorded much higher annual growth of urban population. India, the leading country in South Asia has shown an unprecedented increase in urban population in last few decades and its urban population has increased about 14 fold from 1901 to 2011.

The level and growth of urban population in the recent decades has been concentrated mainly around big cities due to expansion of economic activities and significant increase of service sector. People come from rural area to nearby urban areas in search of livelihood and better living condition.

Out of 10 most populous countries in the world, 6 are in Asian continent, they are China, India, Indonesia, Pakistan, Bangladesh and Japan. Except Japan, majority of people (above 50% of total population) of these countries are living in rural areas. The urban population across the world is growing rapidly and an estimate according to United Nations (UN), nearly 50 million people are added to the world's urban population and about 35 million to the rural population every year. Thus the share of urban population to the world's total population is growing fast and UN estimates that the urban population would be about 66.4% of total population of the world by 2050.

The urbanization is considered as a potential demographic dividend but rapid and unplanned

urbanization is creating havoc, particularly in metropolitan cities in India. The carrying capacity of cities and environmental condition is under threat due to unexpected urban growth and unplanned industrialization respectively. The problem of slums and other infrastructure facilities can be managed through effective infrastructure planning and government initiatives.

2.1 Past and Recent trajectories in India

Indian society is in a stage of urban transition and rural-urban migration has been found to be modest in the country. Among three major components of urban growth (natural increase, net migration and areal reclassification) the largest contributor is the natural growth which contributes about 41.7% during 1971-81 and about 59% during 1981-91 (Bhagat, 1992). This increase complements a corresponding decrease in the percentage share of net migration and reclassification of rural-urban areas. Net migration contributed about 39.4% during 1971-81 which has reduced to about 22.62% during 1981-91. Reclassification of urban area/changes in municipal boundaries contributed 18.85% during 1971-81 which has reduced to about 17.4% during 1981-91.

Copy and paste from SN Nandy
"Urbanisation in India" NOT mentioned
in the references

Copy and paste from Wikipedia "Urbanisation in
India" NOT mentioned in the references

[A RES Project thesis in 2020.]

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BUT mentioned in the references!

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Plagiarism

May be you're not alone



Quelle: www.blick.ch/news/ausland

Karl-Theodor Freiherr zu Guttenberg

16.02.2011

First publications on plagiarism accusations

23.02.2011

Disqualification of the doctorate by the University

01.03.2011

Resignation as german minister of defence



Plagiarism

May be you're not alone



Quelle: www.n-tv.de/politik

Annette Schavan

May 2012

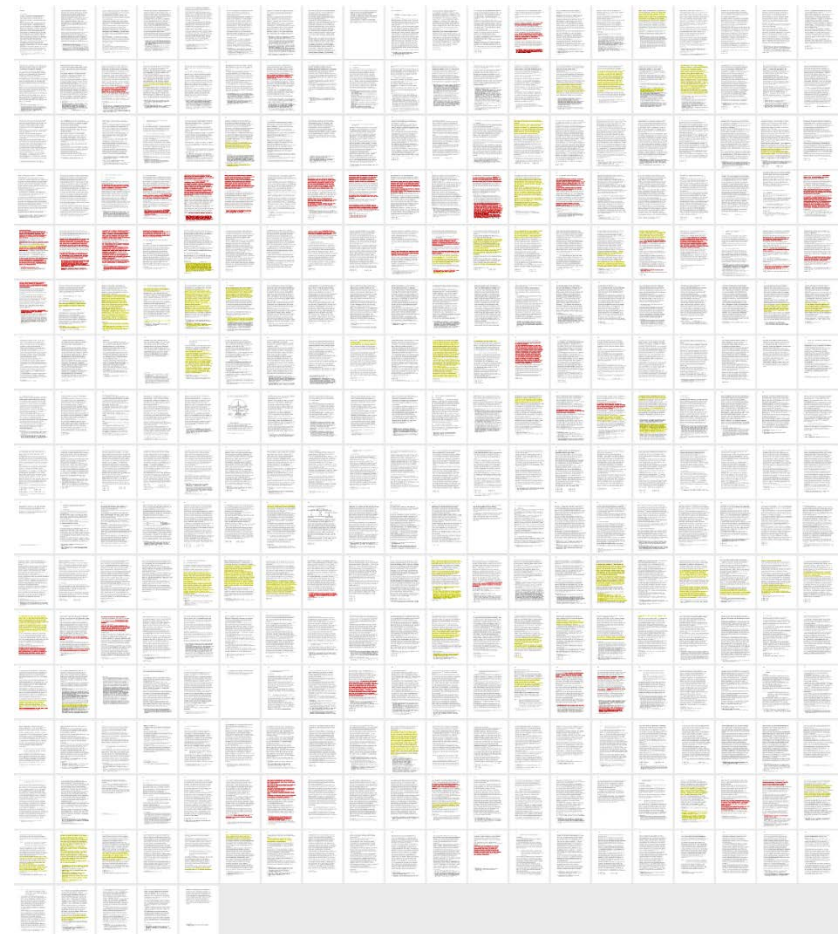
First publications on plagiarism accusations

February 2013

Disqualification of the doctorate by the University

09.02.2013

Resignation as German minister of science



Plagiarism

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Beware! Plagiarism can be detected and the risk of falling is immense.

KI generated Plagiarism

Using ChatGPT or other programs to write your term paper or thesis is in most cases plagiarism! A first draft of future regulations:

1. Brainstorming, structural ideas, first drafts
→ **usage allowed | no citation necessary**, as no citable original sources are available
2. Research and initial references to literature and sources
→ **usage possible | citation of original sources necessary**
3. Summarise state of research
→ **usage not allowed** | core of scientific work
4. Create texts, pictures, graphics, data analysis
→ **usage possible | citation of prompt necessary**
5. Improving language style or translation
→ **usage allowed | no citation necessary**

KI generated Plagiarism

Example for citing a prompt

From the 1990s onwards, photovoltaics played a major role in the public debate as an energy generation technology [1]. It is now an essential part of our energy mix, even if this technology does not yet have the share it could achieve due to various framework conditions [2].

- [1] ChatGPT beta version GPT 3.5-Turbo, output from 28/02/2024 regarding solar energy, prompt: 'Which energy generation technology was discussed the most in the late 1990s?'
- [2] ChatGPT beta version GPT 3.5-Turbo, output from 28/02/2024 regarding solar energy, prompt: 'Which factors are currently inhibiting the further spread of solar energy the most?'